

PSAM 3210

CD Studio: JavaScript

Last Updated: 260911

Program	School of Art, Media, and Technology: Communication Design
CRN	2158
Semester	Fall 2026
Meeting Day	Thursday
Meeting Time	4:00pm - 6:40pm
Building/Room/Zoom	Academic Entrance 63 Fifth Ave Room 622
Instructor & Email	Severin Bunse severinbunse@gmail.com
Class Website	https://severinbunse.github.io/Javascript-2026/

Course Description

This course will focus on extending students' knowledge of programming and computation using the JavaScript language for web and mobile contexts. Students will learn intermediate and advanced methods in JavaScript, the programming language of the web. Students will be introduced to a variety of dynamic use cases, from front-end frameworks and libraries to the creation of desktop and mobile applications.

Course Outline

Part 1: Relearning the web (Weeks 1-3)

Students are re-introduced to basic HTML + CSS concepts and are asked to produce a simple "homepage" site to catalogue the projects for the rest of the semester. If students feel comfortable, they are given the opportunity to integrate javascript into their site. An emphasis is

given on a foundational web knowledge (responsive units, breakpoints, clear code) to provide a framework for more advanced work in the coming weeks.

Part 2: Javascript, iteration, and movement (Weeks 4-6)

Students are introduced to programming concepts: data storage/structures, loops, and functions and explore their visual potential through the development of generative patterns, animations, and modular abstractions. Emphasis is given towards understanding data types, storing values, and manipulating css (units and classes) with javascript.

Part 3: Interaction (Weeks 7-8)

Students are introduced to eventListeners and interactions through a series of quick interface sketches.

Part 4: Fetching (Weeks 9-11)

Students are introduced to javascript as a method of data management. Students are given demos exploring asynchronous functions, APIs, and localStorage to gain an understanding of fetching and pushing data.

Part 5: The world of the web (Weeks 12-15)

After gaining an understanding of how to manipulate css through plain javascript, students are introduced to javascript libraries through a series of small demos. Students are expected to select and become well-acquainted with the documentation of one library through an interface-based project for the remainder of the course.

Learning Outcomes

By the end of the semester, students will be able to:

1. Develop a range of creative artifacts using the JavaScript language
2. Understand the affordances and limitations of using software as a medium
3. Read and understand the source code of both their own and their classmate's projects
4. Independently research and learn unfamiliar technical topics
5. Understand contemporary programming workflows and processes within the JavaScript ecosystem
6. Develop the ability to incorporate external data using Application Programming Interfaces (APIs)

Assessment Criteria

10%	Attendance & Professionalism
5%	Project 1: Hello World
20%	Project 2: Generative Pattern
15%	Project 3: Garden
25%	Project 4: Sender Receiver
25%	Project 5: Library Interface Project
100%	Total

Attendance, Grading and Work Submission Standards, Program Policies, Making Resources, and University Policies

All CD classes adhere to the same program and university policies:

https://docs.google.com/document/d/1u358io8doX_SVVMGqIM_oH5V00lccneYu4Ww-uE55QM/edit?usp=sharing

Projects

Project 1: Hello World

Using your existing knowledge of HTML and CSS (and JavaScript, if you like), create a homepage that represents you in some way. It could introduce yourself, explore an interest or obsession, or give you a chance to realize an idea you've had. Think of it as making a place for yourself on the web.

This site will also be used to catalogue the projects for the rest of the semester. The project is due next class and can develop as we continue.

Project 2: Generative Pattern

You will write a JavaScript program which uses loops and if statements to create a modular generative pattern. You will design at least one single pattern unit (a module) (think: div) that changes over the course of its output.

Your unit should iterate at least 100 times. With each iteration, your unit should somehow modify itself (content or css attributes: size, color, rotation, shape, texture). Your pattern should have at least 2 variables of change (ex: content, color), is allowed to repeat itself (i.e. be a pattern), and is allowed to be as abstract or literal as you want (it can tell a story, or it can be an exercise in form). The goal of this project is to use javascript to dynamically append HTML elements and programmatically alter their attributes. This project will also utilize javascript to explore techniques of animation and interactivity.

Project 3: Garden

Using what you have learned about event listeners, mouse coordinates, and dynamically creating and appending HTML elements, create a virtual garden which implements the ability to render and place elements where you click your mouse.

Project 4: Sender Receiver

Design and develop a website which incorporates an external data feed, using one of the APIs discussed in class. You may also use another API or service that you have researched independently, so long as it provides a live data stream to work with. Consider how your specific design logic might change the meaning of said data, and how the user may (or may not) interact with the feed as they engage with your site.

Project 5: Library Interface Project

Using an API to fetch place-based data and the Three.js library, design and build an interactive 3D globe that catalogues a network of real places connected by a shared purpose — a resource, a service, a community, a cause. Consider how the look of your globe reflects the topic you are working with, and design an interface for how users discover and access the data behind each place.

Materials and Supplies

Laptop

Camera

Software: Git/GitHub, VS Code, Glitch

Schedule

Week	Date	Topic
1	Aug 27	Intro, syllabus Lecture: HTML/CSS refresher Introduction to P1
2	Sep 3	Lecture: JS variables & data types
3	Sep 10	P1 due, critique; introduction to P2 Lecture: For loops, conditionals

Week	Date	Topic
4	Sep 17	Lecture: DOM manipulation – creating elements, changing CSS via JS
5	Sep 24	Lecture setTimeout/setInterval, animation basics;
6	Oct 1	P2 due, critique; intro P3 Lecture: functions & modular code
7	Oct 8	Lecture: eventListeners (click, mousemove, drag)
8	Oct 15	P3 due, critique; intro P4 Lecture: async functions & promises
9	Oct 22	Lecture: Fetch API basics
10	Oct 29	Lecture: Working with APIs, JSON, error handling
11	Nov 5	P4 due, critique; Lecture: localStorage; intro P5
12	Nov 12	Lecture: Three.js fundamentals – scene, camera, geometry, materials; plotting points on a sphere using latitude/longitude
13	Nov 19	Lecture: raycasting – detecting clicks/hover on 3D objects; begin development, individual check-ins
–	Nov 26	No class (Thanksgiving)
14	Dec 3	Development, individual check-ins
15	Dec 10	Final critique & presentation – P5 due

Communication Design Electronic Devices and AI Policy

(From our [shared CD policies](#))

As a Communication Design student, you need to learn the opportunities and limitations in technology, particularly when it comes to emerging tools and interfaces. This can make it difficult to discern when these tools impede our learning.

Please observe the following guidelines and rules regarding devices:

1. Using cell phones or laptops during class for any reason other than coursework is prohibited Parsons-wide. Communicating with people outside of class is a distraction for your classmates. Use digital tools to limit others' access to your attention during class. The following tools are great for protecting your time:

1. [The 'Focus' feature](#) on mac phones and laptops
2. [The 'Foqos' app](#) (free, also can be paired with NFC tags)
3. [The Freedom app](#), which allows for more customization

2. You may not record faculty or other students without their permission, or without an accommodation from the disability office.

And these guidelines regarding AI:

1. You may not use AI to give feedback to other students or to write personal or project statements. These are personal communications, and generic responses are not appropriate.

2. You may not use AI to create content or design documents except in situations where:

1. that exploration is part of the assignment or
2. that exploration is articulated within your concept for the project

Acknowledgments

This course builds on prior PSAM 3210/3030 "CD Studio: JavaScript" syllabi developed by Lukas Eigler-Harding (Fall 2020) and Eric Li (Fall 2024).